10205

C.L.A.S

CLOSED LOOP AGRICULTURAL SYSTEM.

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Abstract

For Egypt's development towards the future, 2030 vision must be achieved by solving its grand challenges. In this project, two grand challenges was addressed and aimed to be solved, the pollution of air, water, an soil, and the low number of agricultural and industrial bases in Egypt. These grand challenges are aimed to be solved by Egypt to not hinder its economy and development. The project is designed to be implemented in STEM High School for Boys – 6th of October as a real life location as it will help in the problem of high cost of food due to the increase in the cost of the fuel by providing a local source of food for the students. Its simple design and mechanism, ability to be held by the user, and the ability to fit at any environment are from its pros. The project is composed of recycled wooden box as an external structure surrounded by a layer of polyethylene, a tray for holding the soil, lettuce plant, ESP32 as a brain for the circuit, power supply to feed the prototype with electricity, sensors for measuring the temperature, soil moisture, humidity, light intensity as parameters. The project met the design requirements as it contains recycled parts, composed of three parameters with their actuators to stabilize the surrounding environment.

Introduction

Egypt struggles with some harsh challenges that hinders its development towards the vision of 2030. Through this vision, Egypt was facing tough problems that may stops the progress of the 2030 vision. One of these problems is the pollution of air, water, and soil, damaging the crops and the crop land. Egypt generates over 100 million tons of solid waste annually including Municipal solid waste, Construction and demolition waste,and Industrial and agricultural waste. Also, Egypt produces 16.4 billion cubic meters of wastewater annually including agricultural sources, and sewage. The health of the soil in the Nile Delta region is declining due to the growing number of enterprises and their emissions, urbanization, increased congestion, and the utilization of sewage and waste sediments. Although the Nile Delta covers only 20,000 km2, it accounts for 46% of Egypt's total agricultural area, which is 55,040 km2

The previous challenges Cause bad consequences, the climate change and the rapid increase in the greenhouse gases in the atmosphere. The climate change occurs because of the burning of the fossil fuels like petroleum. In the last decades, It was observed that the carbon dioxide emissions in Egypt increased to be 258.37 million ton annually as shown in the **Figure 1.**

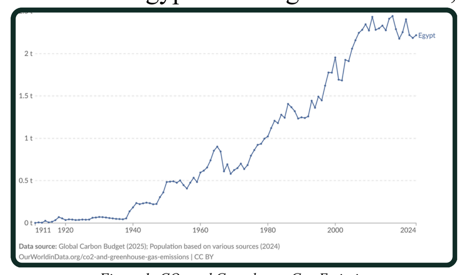


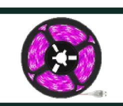
Figure 1: CO_2 and Greenhouse Gas Emissions

Two prior solutions were introduced to address the problem. The first was the automated scale aquaponic system, designed for serve the aquatic life by controlling the fish growth in a neutral environment conditions. Smart agricultural solution to autonmous greenhouse control was the second chosen prior solution. It is designed to control the quantity of air in the vaccum, light, and heat inside the prototype.

During the prototype development process, design requirements were chosen to ensure that the prototype would be effective, eco-friendly, safe, accurate, an precise. The chosen solution was a closed-loop agricultural system for suitable for planting lettuce and pepper. It was selected to be implemented in STEM High School for Boys – 6th of October as a location.

Materials & Methods

Ten essential materials were used in constructing the prototype, ***As shown in Table 1***

Image	Name	Quantity	Description
	Recycled wooden box	1 box	A recycled wooden box to contain all the components in it.
	Polyethylene Film	~ 2 meter-squared	A single layer film used to wrap the prototype to prevent heat leakage.
	Relay module	2 4-channel 1 2-channel	A module which turns on/off the actuators based on a signal from the ESP32.
	ESP-32S 38Pin	1 Dev board	The motherboard of the whole prototype, it is responsible for managing all the parts and the signals passing through.
	Peltier	2 Peltiers	Are devices responsible for generating heat flux by passing electric current to the both sides.
	DC Fan	5 Fans	Are significant for venetrating the prototype parts.
	Growth Light	~ 1 meter	Used to give light to the plant that is responsible for its growth.

Methods

Made the 3D design using onshape CAD software to make sure the dimensions are accurate, ***as shown in Figure 2.***

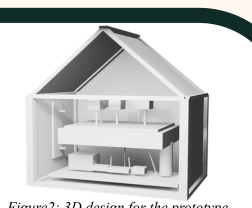


Figure2: 3D design for the prototype

Added all the parts via schematic connection to ensure it is working correctly, ***as shown in Figure 3***.

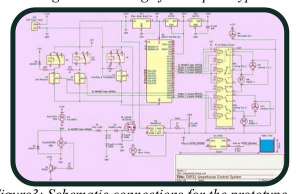


Figure3: Schematic connections for the prototype using kicad

Collected a recycled wooden crate as a body for the prototype, and bought all the electronic parts to build the circuit, ***as shown in Figure 4.***

Build the prototype body using some old wooden parts and the recycled wooden crate by cutting the wood into pieces then sticking them to make the roof of the greenhouse body, ***as shown in Figure 5.***

Figure4: Recycled wooden crate

Added the water container, tray, soil to the greenhouse body as a basic structure, ***as shown in Figure 6.***

Followed the schematic and connected the circuits components correctly, making sure it will work properly.

Implemented the circuit and the rest of the parts of the prototype like the lettuce plant, ***as shown in Figure 7.***

Figure 5: Greeenhouse prototype after circuit implementation

Wrapped the body with a single layer of polyethylene to prevent the leakage of heat from the greenhouse, ***as shown in Figure 7.***

Calibrated the sensors used for the prototype to make sure it is accurate.

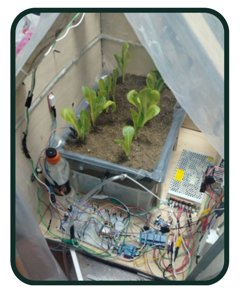


Figure 7: Greeenhouse prototype after circuit implementation and polyethylene wrapping

Figure 6: Greeenhouse prototype after circuit implementation

Test Plan

Test plan The objective of this test plan is to evaluate the validate the prototype and design requirements, focusing on two key aspects, design and mechanism.

Parameters testing

The prototype must be able to measure at least 3 parameters properly, in this case, temperature, soil moisture, and humidity were included in our prototype as parameters.

accuracy

To ensure that the device has good accuracy, the temperature, soil moisture, and humidity are measured using the prototype from the STEM High School for Boys - 6th of October school to make sure it met the design requirements within the selected location.

Results

Negative Results

At first, we brought a stepdown converter to decrease the circiut's voltage and increase the current, but we found that the stepdown was damaged before it was used. This made us took more time than expected to complete the circuit.

Secondly, The power supply that was bought to operate the prototype was 12V and 10A. That was a problem because most of the parts in the circuit needed 7A or more, so this made a significant problem on how to operate all the part with the same power supply. To solve this problem, the components were distributed to the power supply's pins of 10A each to fit all the components.

Positive Results

A reliable closed-loop agricultural system was developed for use in STEM High School for Boys - 6th of October, conforming to the required design requirement, showing the responses from the actuators to the changes in the environment of the project relative to the time taken, and another one showing the environmental parameters changing by passing time.

Table 2: power consumption table

Component	Current (A)	Power (W)
2x Peltiers	10	120
Heating Lamp	4	48
Grow Light	1.5	18
Water Pump	0.6	7.2
5x Fans & Humidifier	0.86	10.4
Sensors	0.01	0.1
TOTAL PEAK	16.97 A	203.7 W

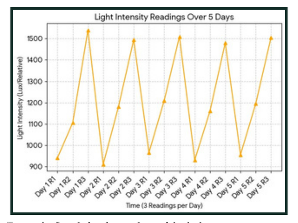


Figure 8: Graph for the reading of the light intinsity over time

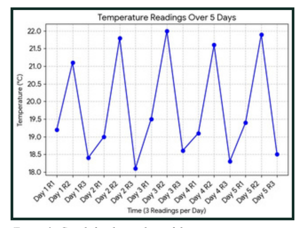


Figure 9: Graph for the reading of the temperature over time

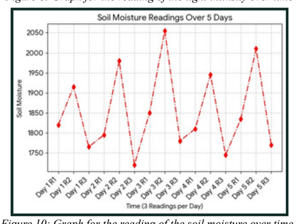


Figure 10: Graph for the reading of the soil moisture over time

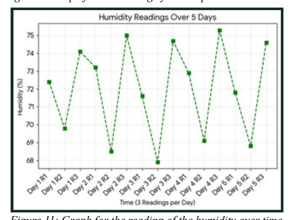


Figure 11: Graph for the reading of the humidity over time

Analysis

Temperature effects on the greenhouse

Temperature is a key factor affecting plant growth, evaporation, and transpiration. In greenhouse systems, solar radiation enters through the transparent covering and is partially trapped, increasing the internal temperature. Research shows that evapotranspiration is one of the most crucial processes controlling water movement in greenhouse crops. As temperature increases, the kinetic energy of water molecules and plant physiological activity also increases, leading to higher transpiration rates and accelerated water loss from the root zone This process is further intensified by increased vapor pressure deficit between the leaf surface and the surrounding air, which enhances the diffusion of water vapor from the plant o the atmosphere. **As shown in Figure 12**, the data collection of the air temperature from the same study mentioned before. The data shows the air temperature over two months, with fractures as the temperature decreases at night and increases during the day, with an average of 22.5 .(Sustain. Food Syst,2023).

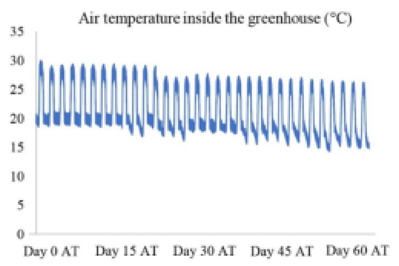


Figure 12: Greenhouse body after adding tray, and cooling system

This is critical for Lettuce and Capsicum annuum, since excessive heat can cause stress and reduce productivity. Moreover, the lettuce needs a temperature between 60 () which can be hard to maintain in the summer due to the temperature. The temperature affects the transpiration rate of the plant as well,following **Equation 1.**

$$E \propto T$$

Equation 1: Relation between energy and temperature

As E is the transpiration rate of the plant and T is the temperature, in other words, the higher the temperature, the greater the loss of water in the plant that the system must maintain.

Soil moisture and irrigation growth model

Soil moisture is an essential topic in green farming, as it indicates when the plants need to be watered. Studies indicate that soil temperature and moisture are strongly interconnected and directly influence plant root growth and water movement. Additionally, higher soil temperature increases evaporation, accelerating moisture loss. However, the soil moisture sensor does not give a true percentage from the beginning; it needs more calculations following **Equation 2**

$$\theta\,(\%) = \frac{R_{dry} - R_{current}}{R_{dry} - R_{wet}} \times 100$$

Equation 2: Relation between energy and temperature

Where θ is the soil moisture by percentage, and the R_dry is the reading of the sensor as the soil is fully dry and R_wet Is the reading when the soil is fully moist with water, and R current is the reading of the sensor when the soil is in its current state.

Light intensity and photosynthesis efficiency

Light intensity is one of the essential factors controlling photosynthesis in a greenhouse. Solar radiation gives the energy required for converting carbon dioxide and water into glucose, directly affecting plant growth and productivity.

Studies indicate that photosynthetic activity is strongly dependent on incident radiation, especially in controlled environments such as greenhouses, where light availability can limit or enhance crop performance. The factors that affect light intensity are distance, following **Equation 3.**

$$I = \frac{P}{4\pi d^2}$$

Equation 3: Relation between light intensity and distance from the light source

I is the light intensity, p is the power of the light source, and d is the distance from the light source. As the plant was moved closer to the grow light by increasing the height, since the distance is squared in the law, the intensity increases quadratically. Increasing light intensity causes an increase in the photosynthesis rate of the plant, following **Equation 4.**

$$P\,(I) = \frac{P_{max} \times I}{K + I}$$

Equation 4: Relation between photosynthesis rate and light intensity

P(I) is the photosynthetic rate and P_max is the maximum photosynthetic rate k is the half-saturation constant; with this law, it can be concluded that at high light intensity (P)≈P_max

Ventilation and roofing impacts on temperature

The greenhouse roof traps heat, increasing internal temperature. Ventilation reduces this heat by allowing air exchange between the inside and outside environments.

Studies indicate that soil temperature and moisture are strongly interconnected and directly influence plant root growth and water movement (Xu Zhang et al., 2025). As soil temperature increases, the thermal energy within the soil rises, enhancing evaporation rates from the surface. At the same time, higher temperatures stimulate root metabolic activity, increasing water uptake by plants. **As shown in Figure 13,** the humidity data collection of the same greenhouse as it was between 50-52% and 90% as it used the fan and the roofing ventilation system in the first graph showing the temp by percentage The combined effect of increased evaporation and root absorption leads to a faster depletion of soil moisture, particularly in greenhouse environments where heat accumulation is significant. The ventilation system contained a 12V fan for ventilation, as decided following **Equation 5.**

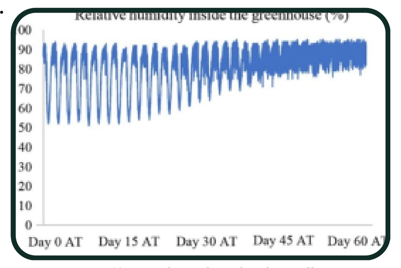


Figure 13: greenhouse humidity data collection

$$Q \propto m \times \Delta T$$

Equation 5: Relation between total heat lost and the air flow rate

Q is the total heat lost, and m is the airflow rate, and is the difference between T inside and T out. Indicating that the heat lost with the ventilation system depends on the strength of the fan used

Table3: Learning outcomes used

Learning Outcome	Concept	Content	Relation
CS.2.06	1	User interface	Helps to understand how to design and manage System using App interface
BI2.09	2 to 4	Hormone Cycle	Helps us to understand how closed feedback system operates
ES.2.09	1 to 6	Earth Cycle	Helps us to understand how closed feedback system operates
PH.2.09	1 to 6	Circuit and charge and discharge	Helps us to understand how closed feedback system operates
PH.2.10	1 to 4	Ac & Dc generator	Helps us to understand how closed feedback system operates
ME.2.05	1	power	Helps us to understand how closed feedback system operates

Conclusions

In conclusion, many grand challenges that hinders its development and its 2030 vision, two main challenges was aimed and addressed by the capstone project, the pollution of air, water, and soil, and the lack in the agricultural and industrial bases. This project aims to solve these problems by providing a solution which is the closed-loop agricultural system, controlling the environmental parameters to make the environmental condition stable. The prototype was build from recycled materials, except the electronic components. It was chosen to integrate the project in STEM High School for Boys – 6th of October as a real-life location, solving the problem of high budget for buying and delivering the food to the students by integrate the prototype in the school to produce its own food, saving its budget.

Recommendation

Real-Life Location

Boarding schools face multiple difficulties regarding funds. There has been a rapid increase in services and products, which makes it a necessity for schools to find an alternative. **As shown in Figure 14,** it is highly recommended to make a real-life application of the project in STEM High School for Boys – 6th of October as the location for the implementation of our device's application.

At present, our school depends on external suppliers and delivery services which increase the costs on the school's financial manager, increasing the funds that the school apply for. This adds more pressure on it to cover all the students' demands during their learning journey. This solution will help the school to save on the delivery and ship the food that comes to the students every day by using the project to produce its own demand from fruits and vegetables without needing any shipping which cases a lot. Giving all the students their specified portions of food without any decrease in its amount.

Figure 14: Real-Life Location

Solar Photovoltaic Panels Integration

A highly promising direction in the future development of the project is to integrate a solar photovoltaic panel with energy buffer units in the project to ensure energy continuity. A solar photovoltaic panel is a device that converts the sunlight into electric energy **as shown in Figure 15**, will be coupled with a buffer unit which acts as an energy storage that is settled between the solar panel and the site of the electrical load. When the panels collect the sunlight and convert it into current, this current will be used in the prototype during it is working, while the excess electricity will be stored in the buffer for future using during night or cloudy weather, helping in increasing the project efficiency, and using a source of renewable energy which is the sunlight. This makes sure that the prototype will be working with no obstacles in energy. However, it is not implemented due to difficulties regarding its high initial cost.

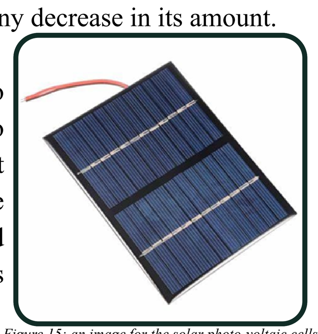


Figure 15: an image for the solar photo-voltaic cells

Polycarbonate Implementation

It is also highly recommended to implement polycarbonate sheets instead of polyethylene single layer sheets **as shown in Figure 16**, as it will help in developing and improving the prototype efficiency. Polycarbonate is a high-performance, transparent thermoplastic known for its exceptional durability, light weight, and excellent optical clarity. These properties make it one of the most widely used alternatives to glass in both architectural and industrial applications. There are many differences between the polycarbonate and polyethylene sheets, making it more favourable to be implemented in the real-life project, such as High impact resistance, Superior thermal insulation, Excellent light transmission, Built-in UV protection, Long lifespan (10–25 years), and Reduces energy and heating costs. It is not implemented in our prototype because, it costs more than expected to fund it.

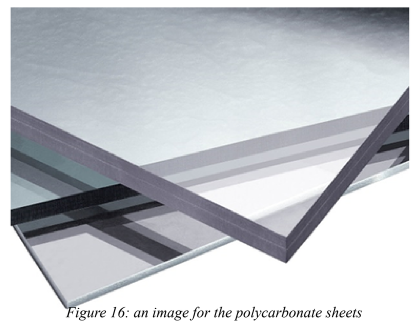


Figure 16: an image for the polycarbonate sheets

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